Yunkang Cheng | 程允康

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Department of Automation, Tsinghua University, Beijing, China

EDUCATION

Tsinghua University, Beijing, China **Aug. 2024 - 2029 (expected)**

B.Eng. in Automation (in progress), Department of Automation

- Cumulative GPA: 3.7/4.0.
- Selected coursework: Linear Algebra (A-), Calculus A(1) (A-), Object-Oriented Programming Practicum (A), Digital Electronics Laboratory (A-), Electronics Design Project (A-).

INTERNSHIP EXPERIENCE

Booster Robotics, Beijing, China **Jun. 2026 - Present**

Motion Control Algorithm Engineer (Intern), Booster Lab

RESEARCH EXPERIENCE

Learning Roller-Skating Motions of Humanoid Robots Based on Adversarial Motion Priors

https://arxiv.org/abs/2607.10815 | Project website

Yunkang Cheng*, *Yutong Wu*, Menghan Li, Shihe Zhou, and Mingguo Zhao*

CLAWAR 2026, oral presentation. * Equal contribution.

- Trained AMP-based policies from human demonstrations for two roller-skating gaits, and demonstrated both on Booster T1.
- My contribution: led the project and policy development; contributed to wheel-ground contact modeling and sim-to-real deployment.

Humanoid Robot Soccer, Tsinghua Hephaestus **2025 - 2026**

Core Team Member | Advisor: Prof. Mingguo Zhao

- Developed obstacle-avoidance and navigation algorithms for humanoid soccer, including high-speed motion, field re-entry, and positioning during matches.
- Integrated and tested navigation behaviors with teammates during preparation and on-site deployment for RoboCup and the World Humanoid Robot Games.

SKILLS AND INTERESTS

Programming: Python, C/C++, MATLAB.

Tools and platforms: PyTorch, Isaac Lab, Isaac Gym, MuJoCo; Booster T1 and Qmini.

Research interests: Robot learning, reinforcement and imitation learning, humanoid locomotion, contact dynamics, and sim-to-real transfer.

SELECTED ENGINEERING PROJECTS

Passive Dynamic Walking: Modeling and Stability Analysis **Oct. 2026**

Course Project in Humanoid Robotics, Tsinghua University

- Modeled a passive biped with circular-arc feet in MATLAB using hybrid dynamics and impact maps; studied the effects of mass, center-of-mass location, and spring stiffness on gait.
- Computed fixed points of the Poincaré return map and assessed local stability using the spectral radius of its Jacobian; evaluated perturbation recovery over 50 steps for 48 parameter configurations.

Qmini Bipedal Robot: Assembly and Integration **Fall 2025**

Course Project in Manufacturing Engineering, Tsinghua University

- Assembled and modified a 12-DoF Qmini biped; resolved mechanical-fit and wiring issues, calibrated its standing posture, and tested motion-control functions.

HONORS AND AWARDS

Competition honors listed below were awarded to teams.

RoboCup, Humanoid Soccer League - Large Division **2026**

Champion, Tsinghua Hephaestus; Incheon, Republic of Korea.

World Humanoid Robot Games, 5v5 Soccer **2025, 2026**

Champion with Tsinghua Hephaestus; Large Robots category in 2026.

RoboCup China Open, Humanoid League (Large) **2026**

First Prize in soccer; Champion of the Technical Challenge.

Tsinghua Future Robotics Challenge **2026**

First Prize; Team Captain.

Second Embodied Intelligence Challenge, Department of Automation **2025**

First Prize; Deputy Team Captain.

National Encouragement Scholarship, Tsinghua University **2026**

Also received the Tsinghua University Comprehensive Excellence Award.

Spark Fellowship, Tsinghua University **2026**

Selected for the 20th cohort of Tsinghua's undergraduate research and innovation program.

Tang Chung Ying Moral Education Scholarship, Tsinghua University **2024**

EXTRACURRICULAR EXPERIENCE

Tang Chung Ying Volunteer Association, Tsinghua University **Feb. 2026 - Present**

Vice President for General Affairs (since May 2026); previously Head of General Affairs

- Coordinate association operations and volunteer recruitment.

Embodied Intelligence Industry Field Study, Guangdong **Aug. 2026**

Team Leader, Department of Automation

- Led a 17-member team on an industry field study in Shenzhen and Guangzhou.
- Team award: Gold Award for Social Practice, Tsinghua University (the department's third such award).

Rural Revitalization and Education Outreach, Fangzheng **Jan. 2026**

Team Leader, Tsinghua University

- Led a 15-member team in robotics outreach and helped launch an online teaching program.
- Team awards for social practice: Silver Award (university level); Gold Award (School of Architecture).